

Semester Use Case — Remote Patient Monitoring for Home Peritoneal Dialysis

For the semester I am selecting **remote patient monitoring for home peritoneal dialysis (PD-RPM)** — a home-based capability that monitors adults with end-stage kidney disease performing peritoneal dialysis at home and escalates to the home dialysis care team when the data indicate the therapy is failing.

Three reasons for this use case rather than a more common selection such as heart failure or hypertension monitoring.

The policy and the outcomes point in opposite directions. The 2019 Advancing American Kidney Health initiative and CMS's ESRD Treatment Choices model both push hard toward home dialysis. But home PD carries a persistent technique-failure problem, with peritonitis and inadequate ultrafiltration among the leading reasons patients transfer back to in-center hemodialysis. We are actively steering patients toward a modality whose retention problem we have not solved. That is a systems gap, not a clinical-knowledge gap.

The signal already exists and is being discarded. A PD cycler generates a complete record every night — fill and drain volumes, dwell times, net ultrafiltration, alarm history. Add patient-reported weight, blood pressure, and exit-site condition, and the trajectory toward technique failure becomes visible days to weeks in advance. Much of that data is currently reviewed only at a monthly clinic visit. This is close to a textbook learning-horizon failure: the information exists, but the consequence lands outside the interval in which anyone is looking.

It has real boundary complexity. The most interesting decision in building the context diagram was that the **PD cycler is outside the system boundary**. It is a separately regulated Class II device with its own manufacturer and control loop. PD-RPM consumes its telemetry and pushes prescription updates across a defined interface, but never delivers therapy. Home hemodialysis is also excluded, since water treatment and vascular access are whole subsystems that would make the scope unbounded. Stating those exclusions explicitly proved more clarifying than enumerating what sits inside.

The system spans all three environment tiers introduced in this module: home-based care as the primary setting, a clinical monitoring node at the dialysis organization, and urgent in-center or emergency escalation at the limit.